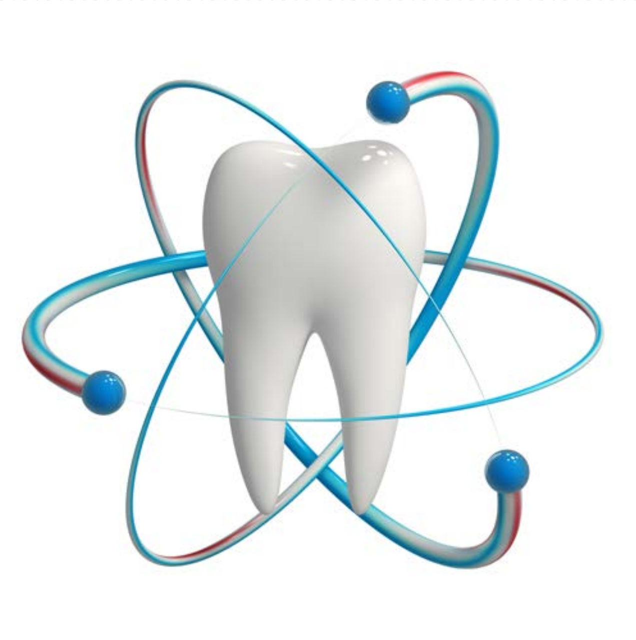
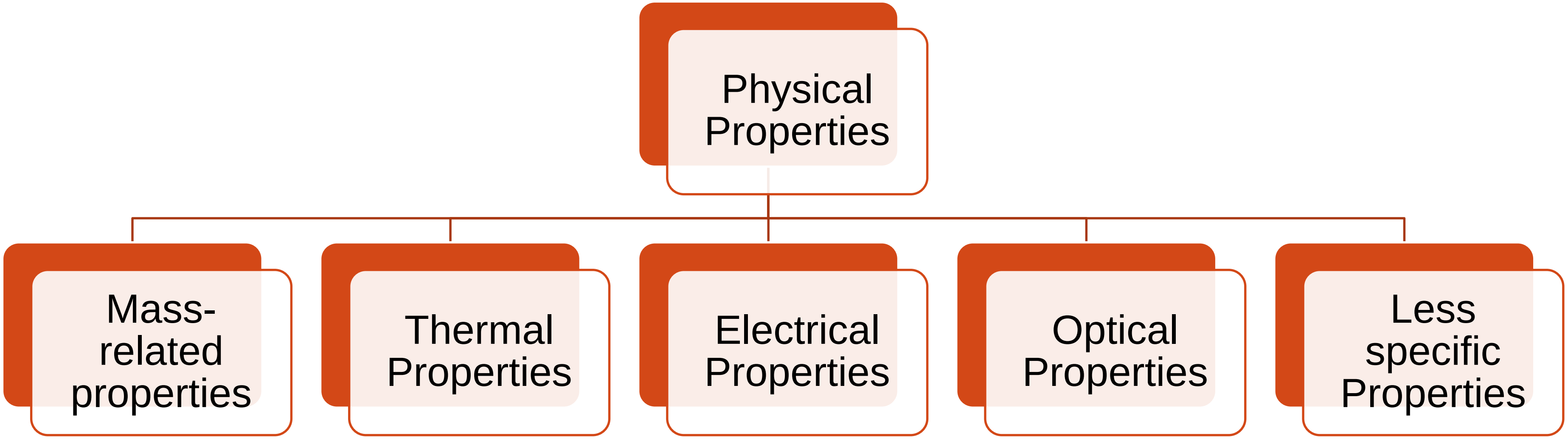
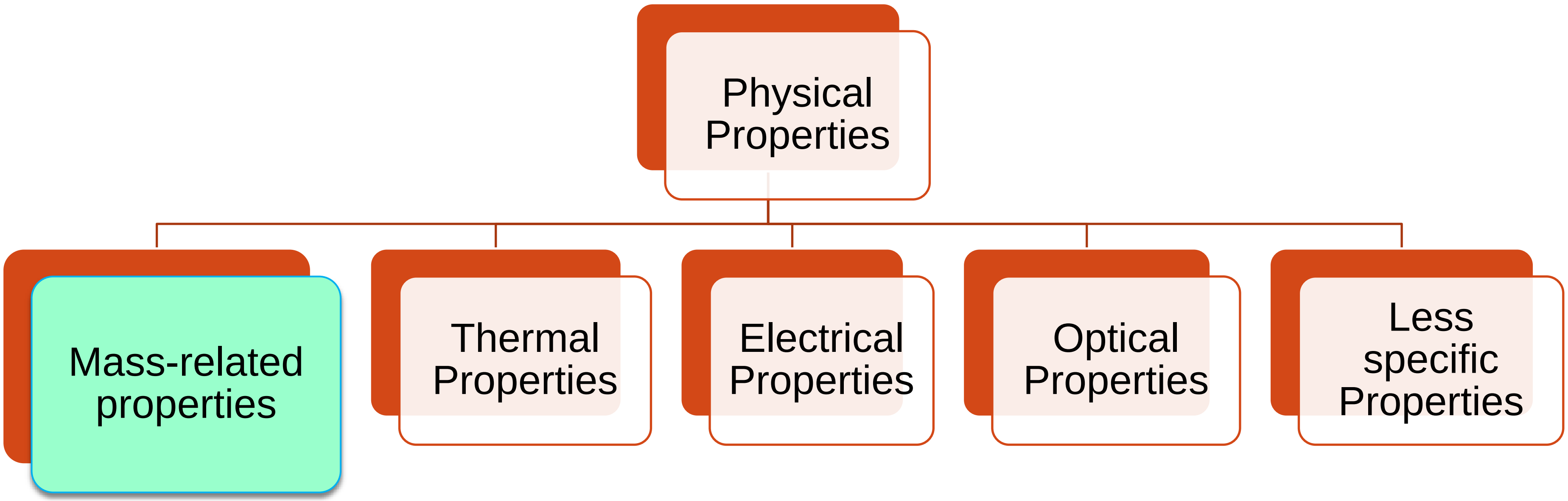
Physical Properties



PHYSICAL PROPERTIES

They are properties that are not related to application of force to the body


Physical Properties
Mass-related properties
Thermal Properties
Electrical Properties
Optical Properties
Less specific Properties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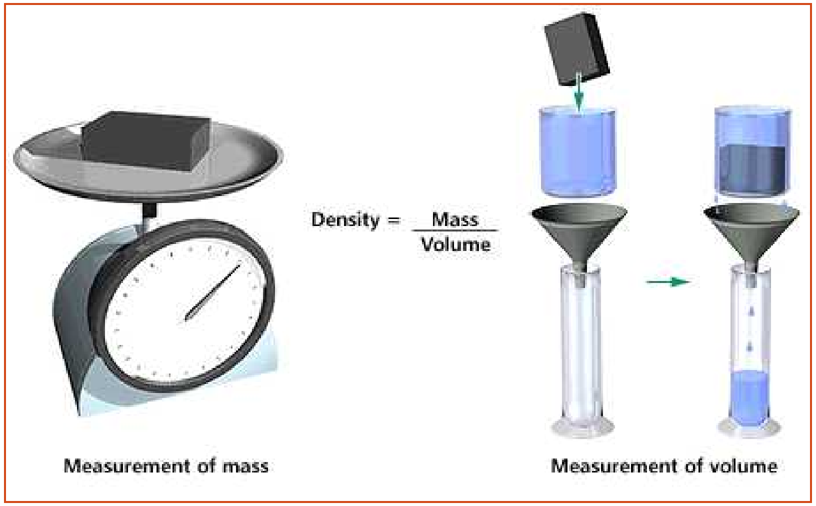
Density & specific Gravity

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

or, in short form:

$$d = \frac{m}{v} \quad \mathrm{g/cm^3}$$



$$\text{Specific gravity} = \frac{\text{Density of material}}{\text{Density of water at } 4^\circ\text{ C}}$$

Importance in Dentistry

- 1) Low Density for the retention of the upper denture and patient’s comfort

Complete Denture:

Components of complete denture:

- **Artificial teeth**

→Porcelain
→Acrylic Resin

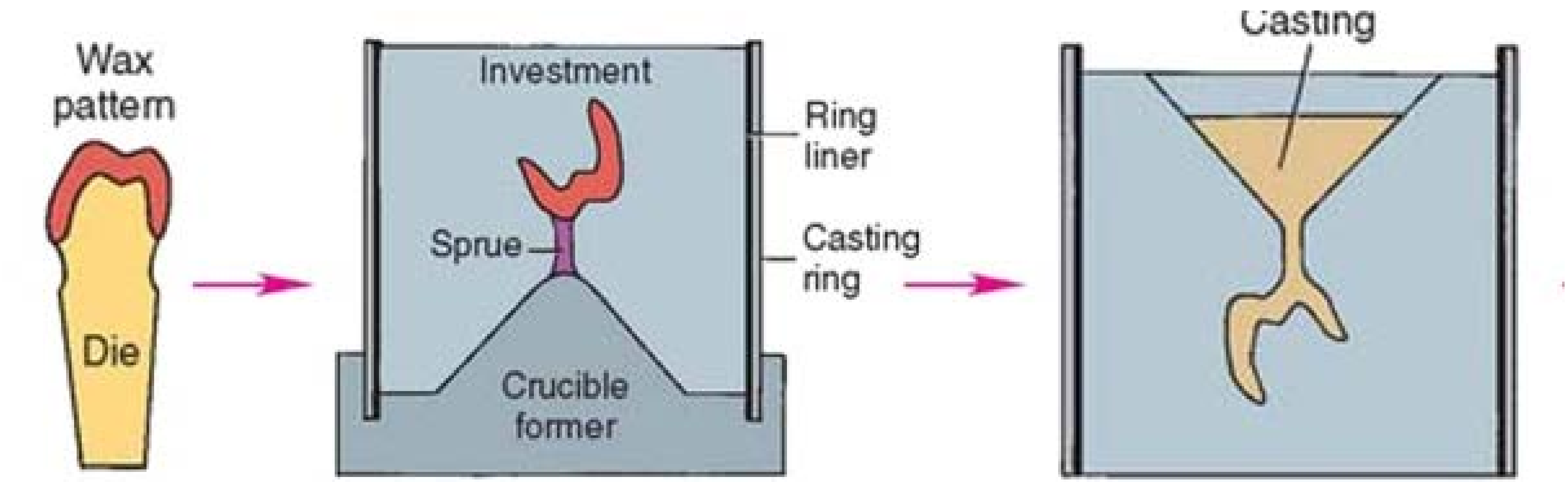
Denture Base

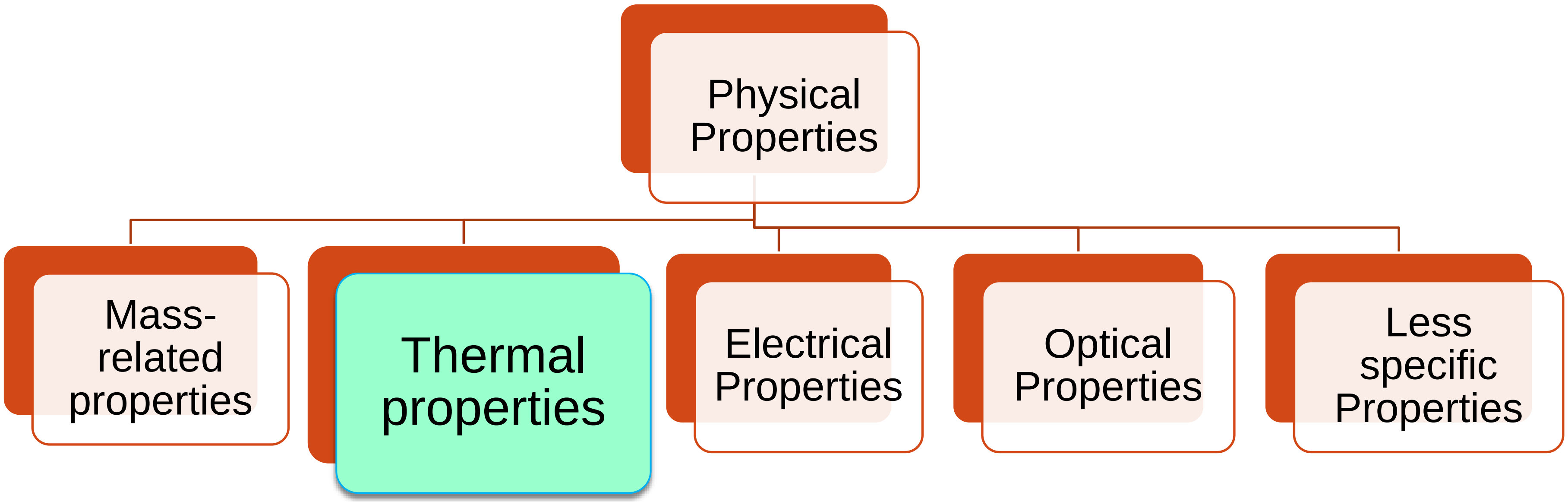
→Metallic(Co-Cr alloy-Gold)
→Non Metallic (Acrylic Resin)

- Non metals < Metals
- Base metals < Gold

2) High Density for better castability (rapid and complete filling of the mold)

- **Gold > Base metals**
- Low density alloys require more casting force


Wax
pattern
Die
Investment
Sprue
Crucible
former
Ring
liner
Casting
ring
Casting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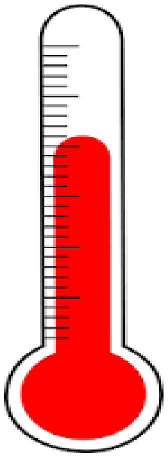
- 1- Specific Heat
- 2- Melting & Freezing Temperature
- 3- Heat of fusion
- 4- Thermal Conductivity
- 5-Thermal Diffusivity
- 6- Coefficient of thermal expansion & contraction (α)

1- Specific Heat:

The specific heat of a substance is the quantity of heat needed to raise the temperature of a unit mass (one gram) of the *<u>substance</u>* 1ºC.



Heat Capacity

- It is the quantity of heat required to raise the temperature of the ***<u>body</u>*** 1°C

Heat capacity =
Specific heat x mass of the body

Clinical significance

Heat is applied to metals during casting to raise the temperature to the melting point

Low specific heat ⟶ No need for prolonged heating

2- Melting & Freezing Temperature

- It is the temperature at which the material melts or freezes

Importance in Dentistry:

- To determine the melting machine used in casting.

- **To determine the type of mold material (investment material) into which the alloy is cast**

- **To avoid over-heating that may lead to..**

3- Heat of fusion:

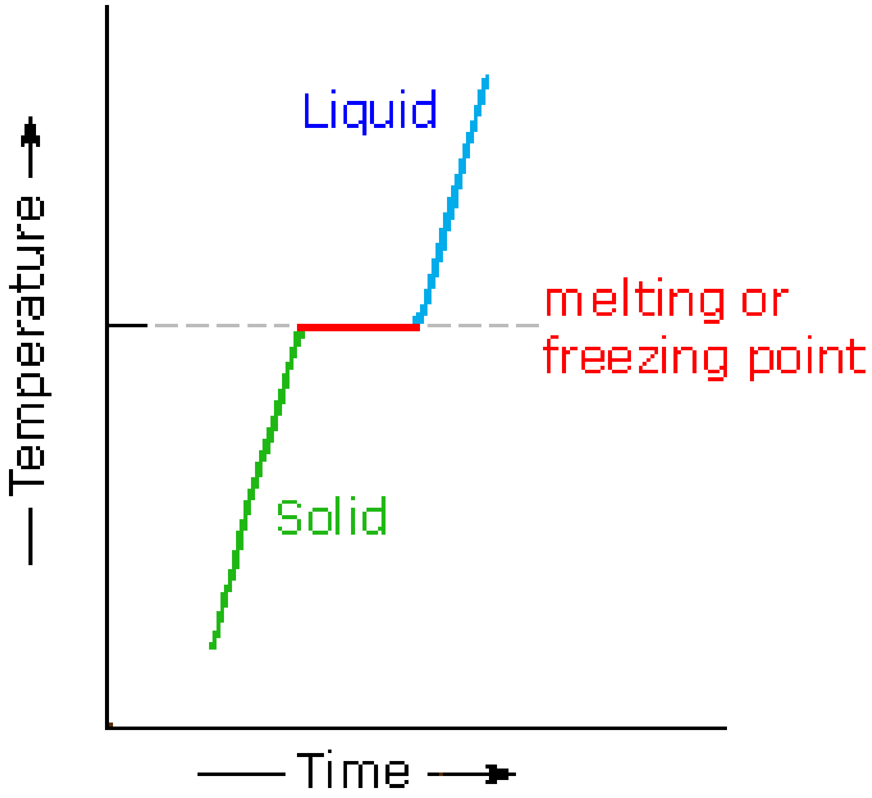
It is the amount of heat in calories required to convert one gram of a material from the solid state to the liquid state <u>at the melting point</u>.

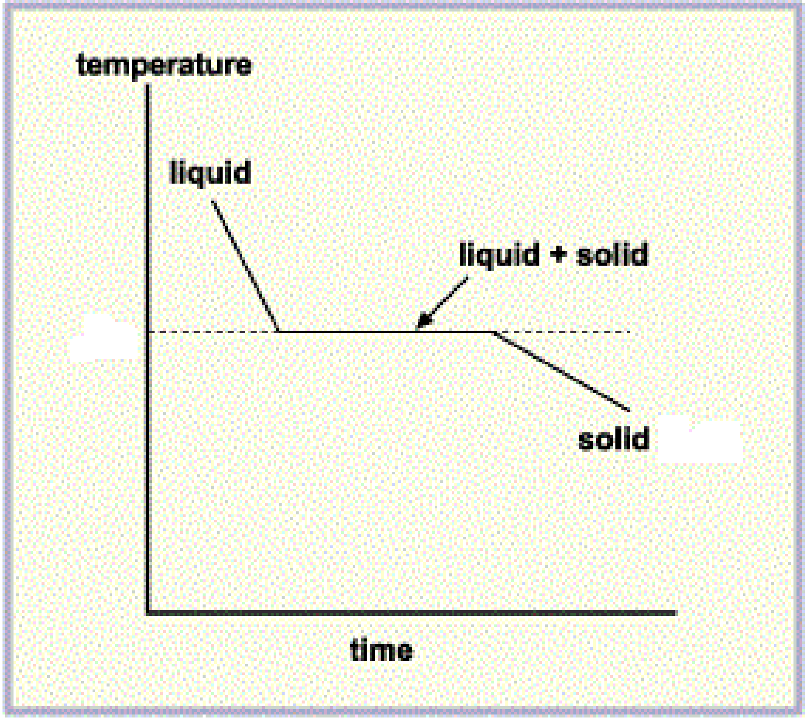
<u>Latent heat of fusion:</u>

It is the amount of heat in calories *liberated* during conversion of one gram of a material from the liquid state to the solid state <u>at the freezing point.</u>

Heat of fusion

Latent heat of fusion





Dental significance

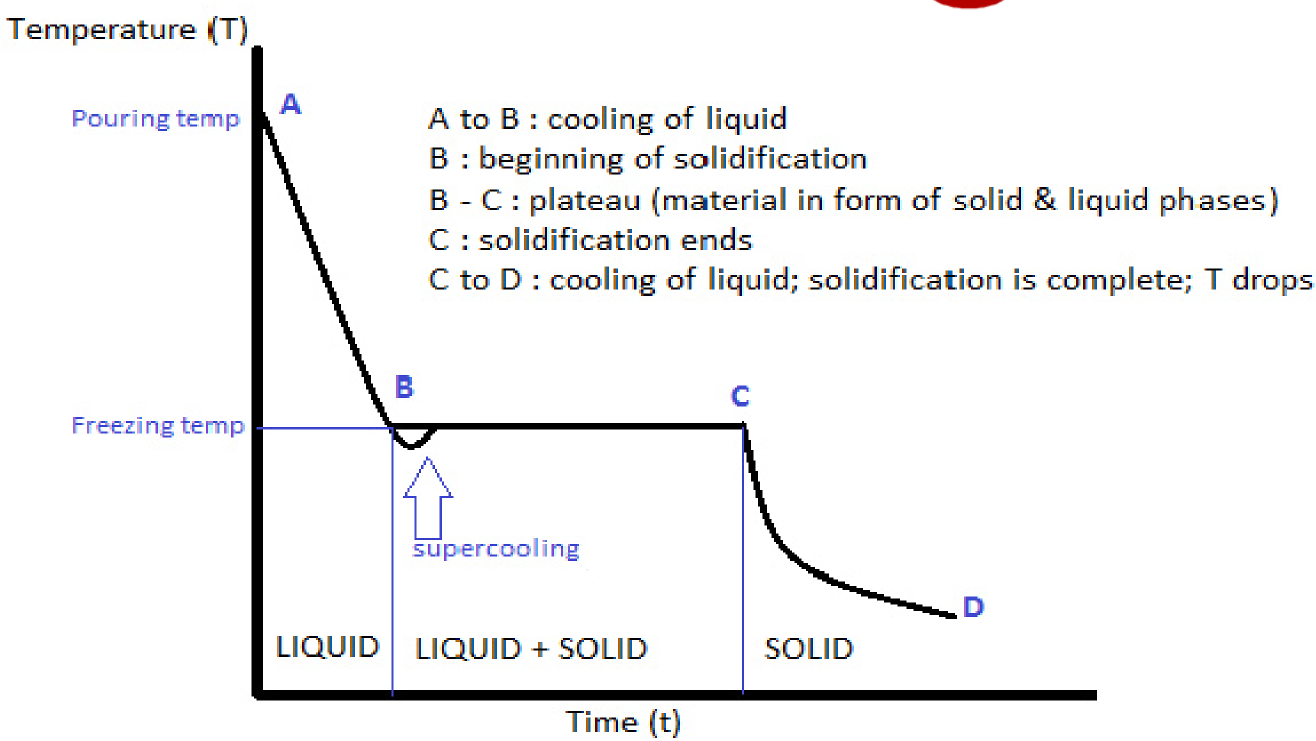


Diagram showing cooling curve of a pure metal

- During solidification, the temperature of the solidifying liquid does not change with time due to liberation of latent heat of fusion (which was retained/ stored by the liquid)

Dental significance….(cont)

During casting, metal must be heated 100°C above its melting temperature to avoid incomplete casting & to allow the molten alloy to completely fill the mold cavity without early solidification.

4- Thermal Conductivity:

It is the quantity of heat in calories per second passing through a body 1 cm thick with a cross section of 1 cm^2 when the temperature difference is 1ºC.

Its unit: cal/sec/cm^2/ºC/cm

Clinical significance

In deep cavities (insufficient dentin thickness over the pulp), large metallic fillings may lead to thermal irritation (patient discomfort-thermal pulp shock.

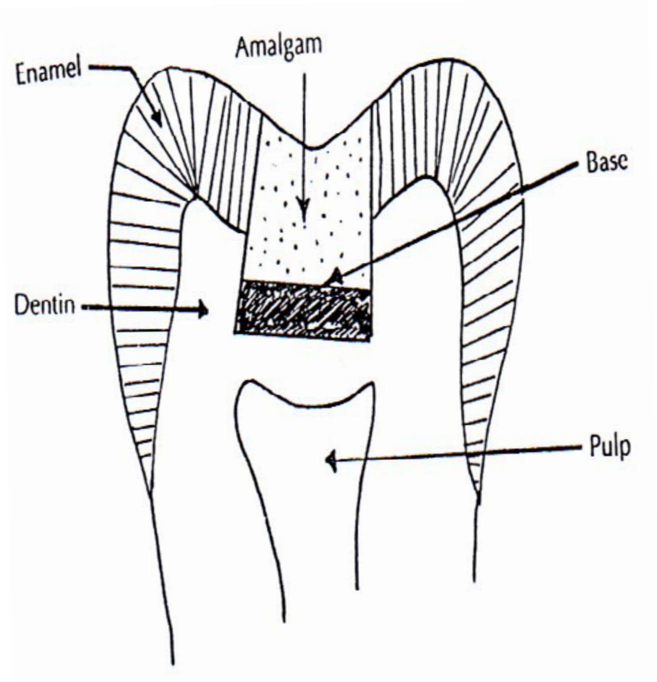
Low specific heat and high thermal conductivity????

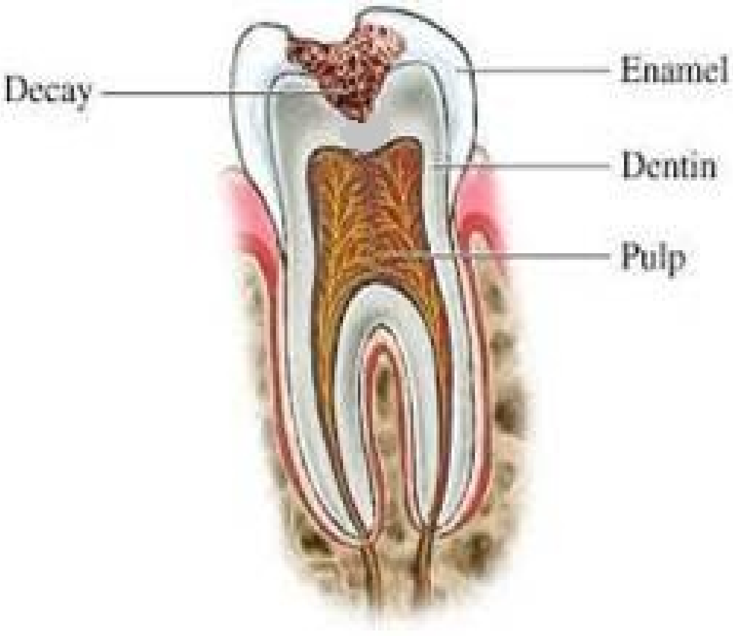
- ∴ *an insulating cement base (non metal) is placed under the restoration*
- ***Base should has:***

High specific heat and sufficient thickness

with

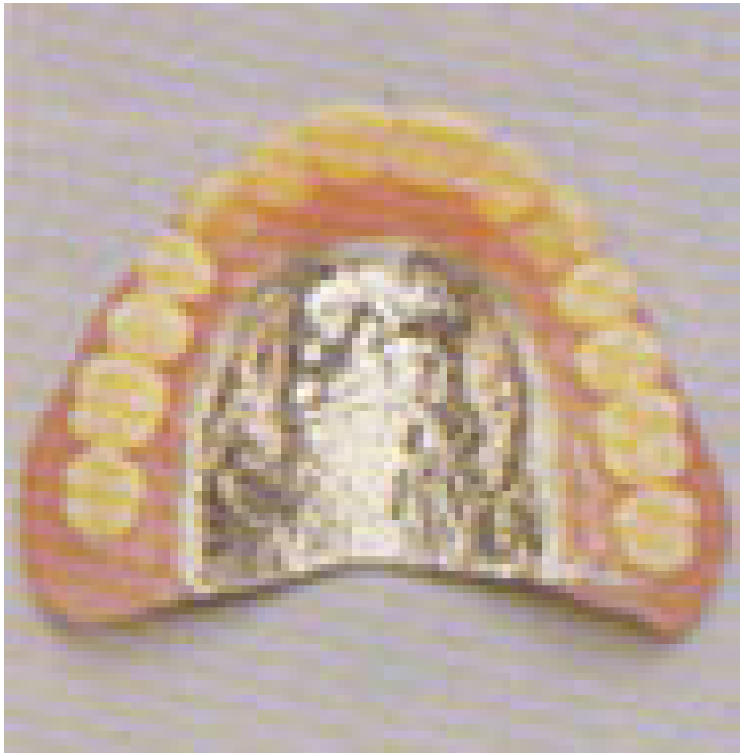
Low thermal conductivity and diffusivity





Dental significance….(cont)

High thermal conductivity of metallic denture base is an advantage …..to produce thermal stimulation to the supporting soft tissue



5. Thermal Diffusivity:

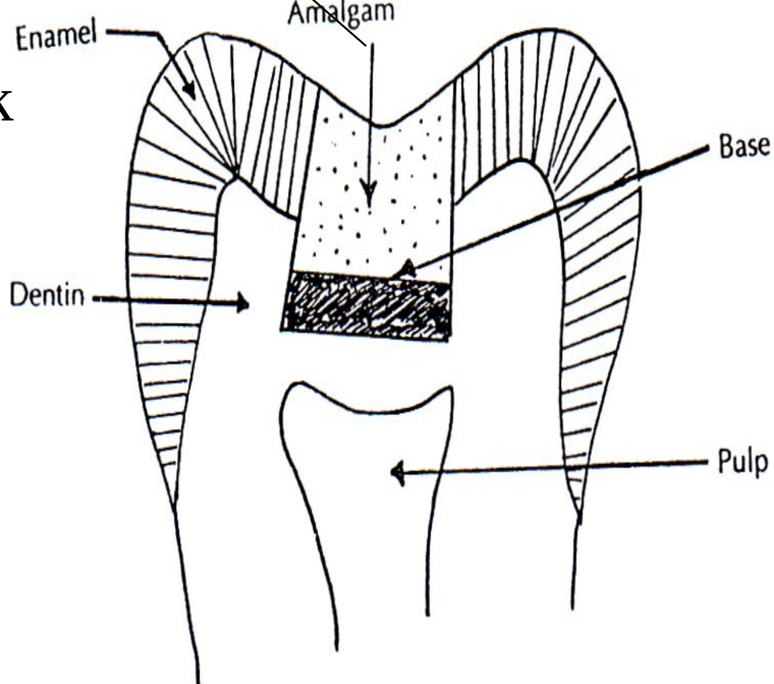
It is the rate of diffusion of heat in the body (it is a measure of transient heat flow)

$$\text{Thermal Diffusivity} = \frac{\text{Thermal conductivity}}{\text{Specific heat x Density}}$$

Unit: mm^2/sec

Clinical significance

Low specific heat + high thermal conductivity of amalgam =thermal shock



Insulating efficiency depends on thermal diffusivity and thickness of the cement base

6- Coefficient of thermal expansion & contraction (α):

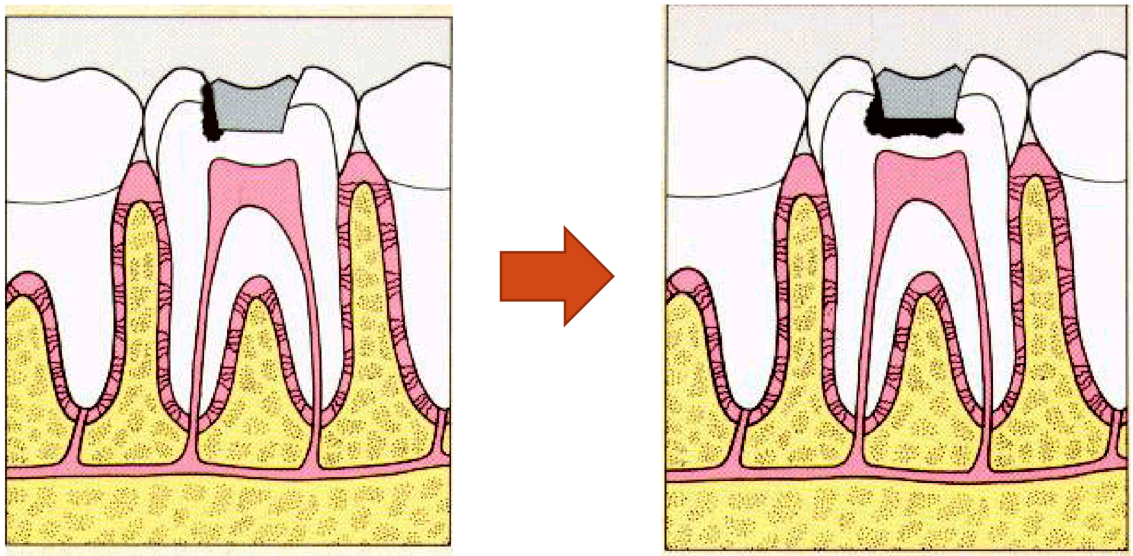
It is the change in length per unit length of a material for a 1ºC change in temperature

$$\alpha = \frac{L\ \text{final} - L\ \text{original}}{L_o\ ({}^{o}C_f - {}^{o}C_o)}$$

Unit: /ºC

or ppm (part per million)

Clinical significance



Difference in α between tooth structure and filling leads to ***marginal leakage***

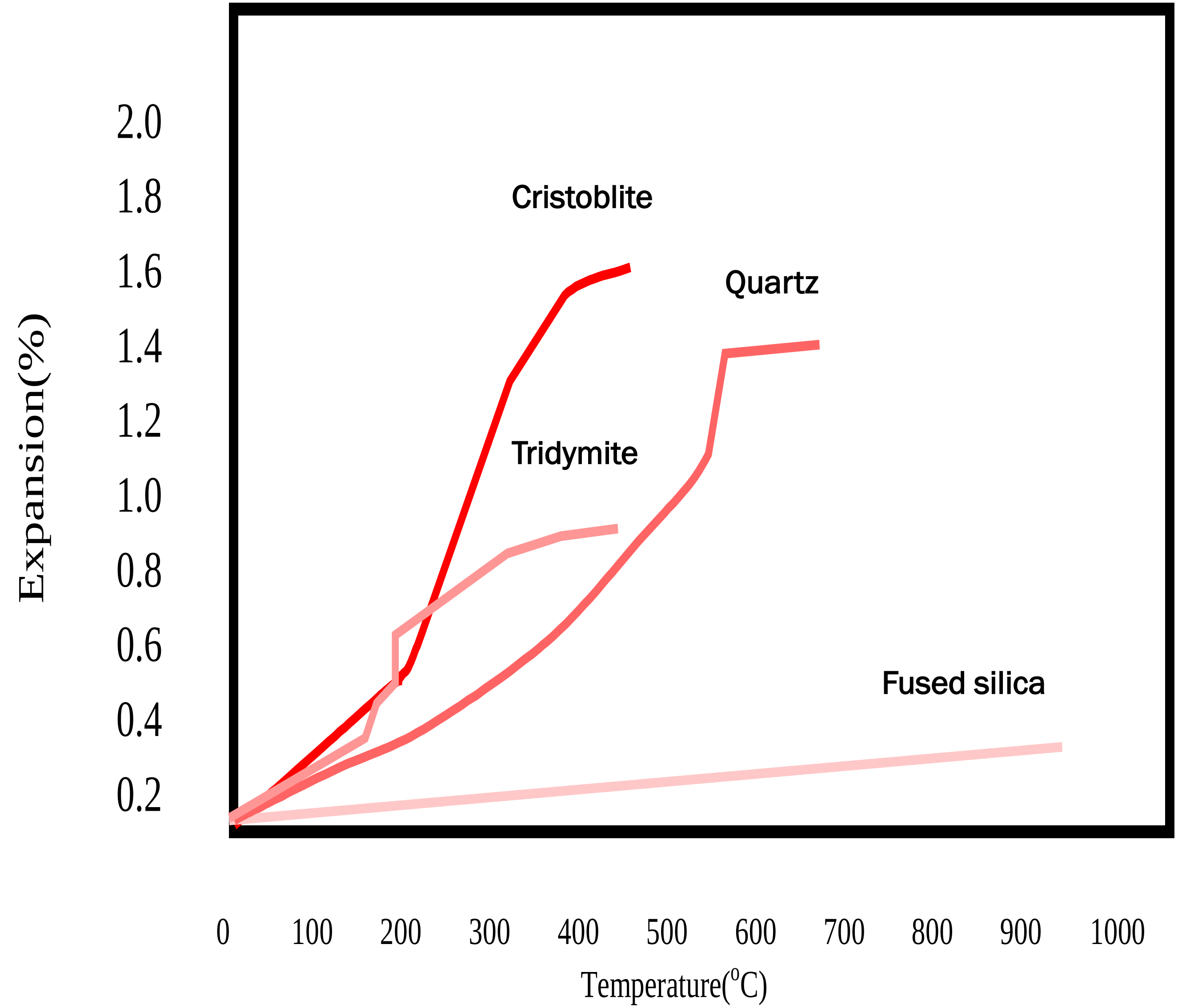
N.B. Marginal Percolation ≠ marginal leakage

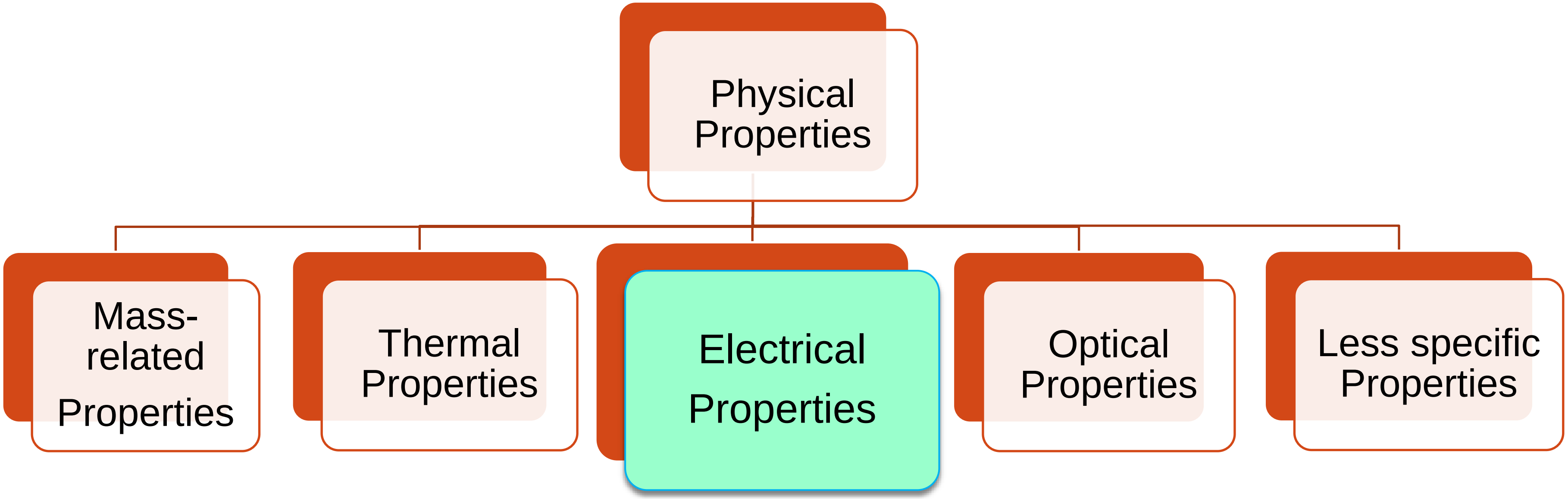
Matching of α between metal and ceramic (porcelain) in ceramo-metallic restoration

Matching of α between artificial teeth and denture base to avoid crazing at the necks of the teeth

Wax has a high **α**, so it should not be exposed to temperature changes before casting procedure to avoid distortion

Thermal expansion by quartz and crystobalite due to displasive transformation





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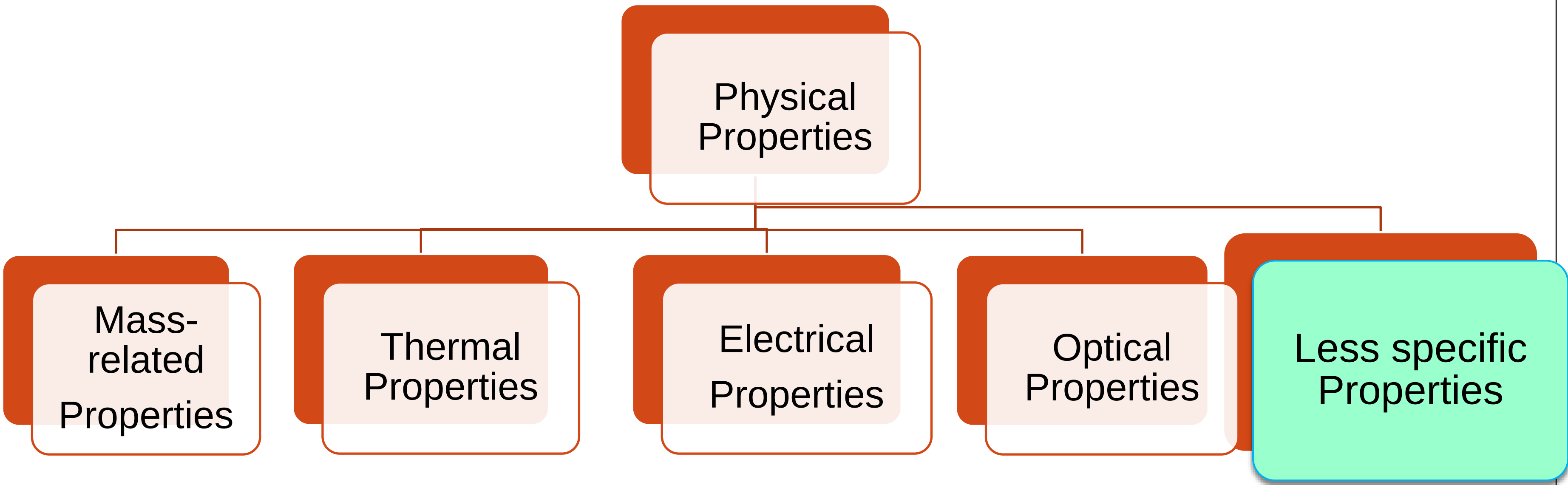
Electrical Conductivity and Electrical Resistivity

Electrical conductivity is the ability of the material to conduct an electric current

- <u>Importance:</u>

- Electrical resistivity of cement base is important (glass ionomer < zinc phosphate & zinc polycarboxylate < Zinc oxide eugenol)
- Carious tooth has less resistivity than sound tooth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Water Sorption

It represents the amount of water adsorbed on the surface and absorbed into the body of the material

IMPORTANCE

Plastic denture base material (Acrylic Resin) have the tendency for water sorption:

- High % of H_2O sorption → dimensional changes & Warpage

- Small amount of H_2O sorption → compensate **Thermal shrinkage**

Importance

The tendency for hydrocolloid impression for water sorption if immersed in water → dimensional changes

Fluidity, Viscosity & Plasticity

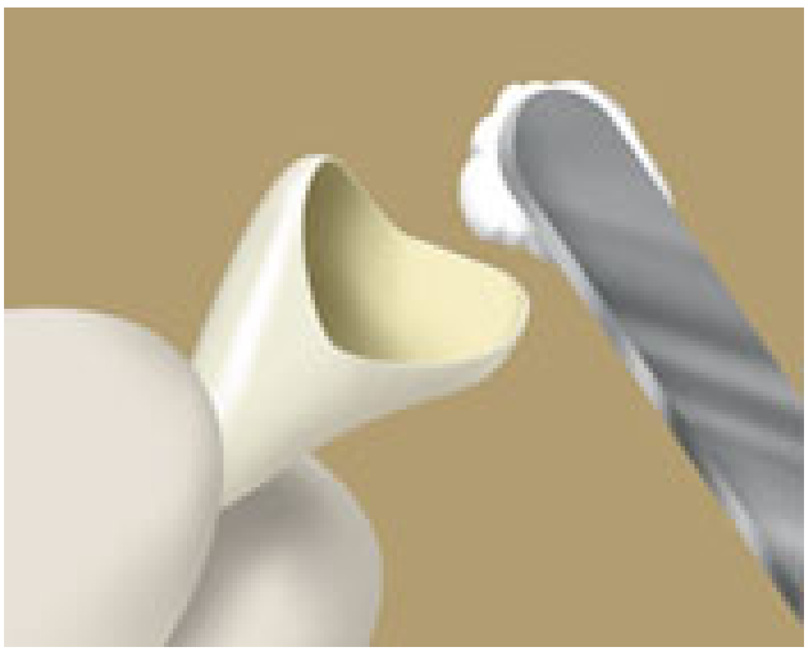
- **Fluidity is the tendency of liquids to flow**

- **While viscosity is the resistance to flow**

- **Plasticity is related to solids or semisolids (the material to be permanently deformed under force (Example: Waxes)**

Importance:

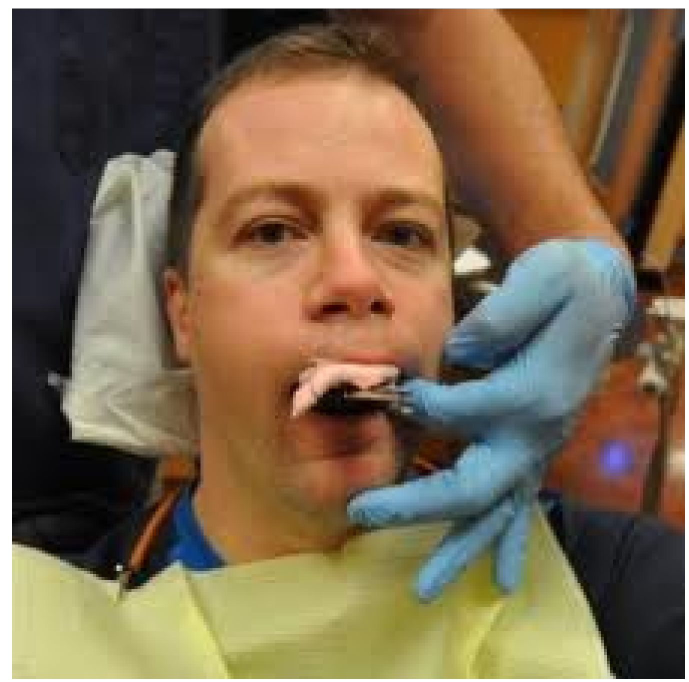
Fluidity & Viscosity play role in penetration, diffusion & wetting ⟶ important in Adhesive Dentistry



Importance:

Most dental materials are manipulated while in fluid/viscous state

Eg: Proper flow of impression materials is essential for dimensional accuracy



Shelf Life

- It describes the general deterioration and change in quality of materials during shipment and storage


Thank You